"Greedy Method"

We can get MST without finding the number of Spanning Trees

Prim's Algorithm for MST

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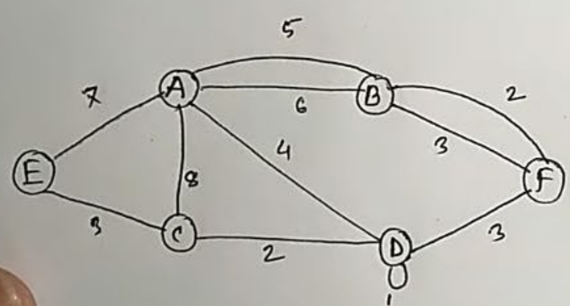
(i) Graph থেকে Loop delete করতে হবে।

(ii) Graph থেকে Parallel edge delete করতে হবে।

(iii) cycle হওয়া যাবে না, নতুন MST এর জন্য।



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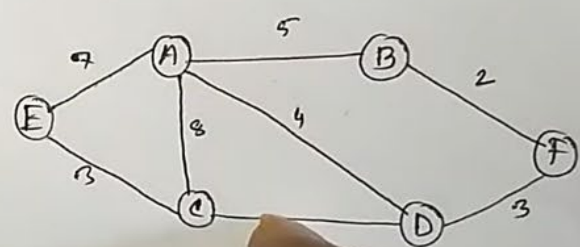
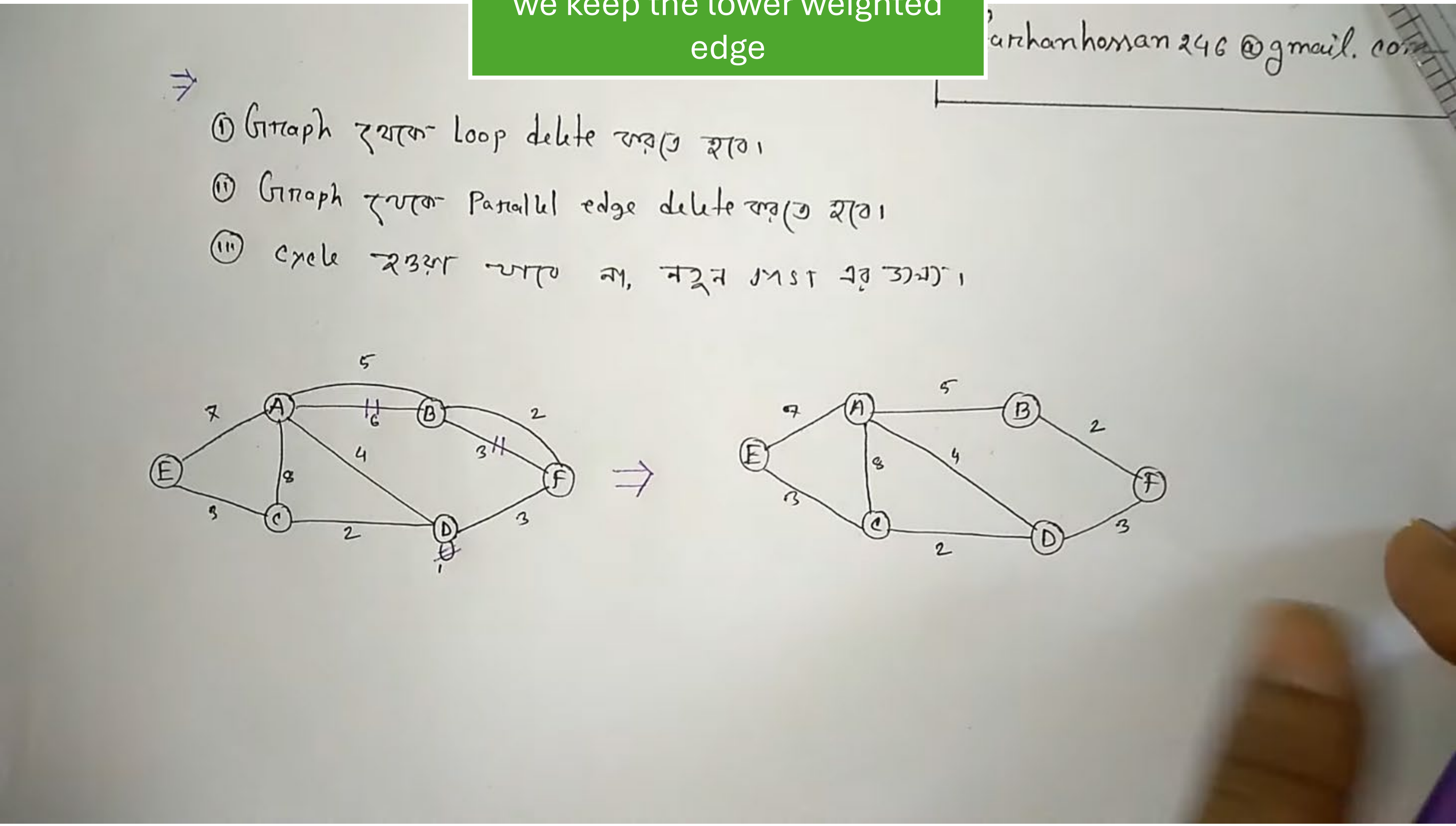


When deleting parallel edges, we keep the lower weighted edge

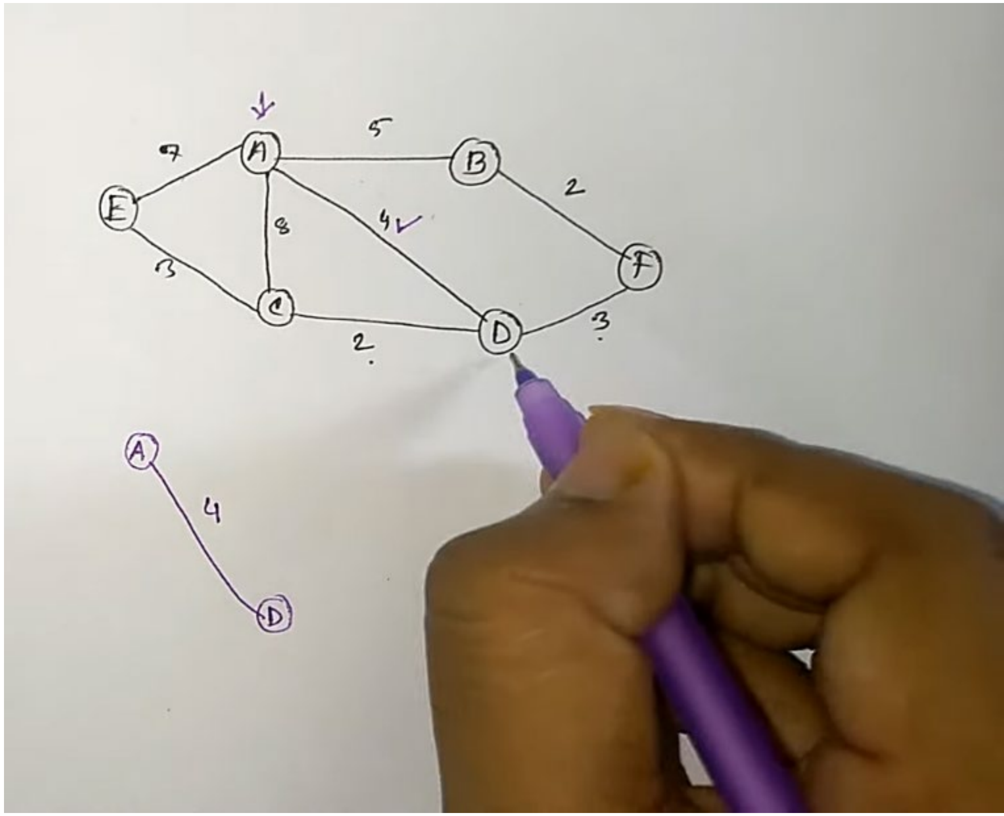


Step 1

For prim's algorithm, we arbitrarily start from any vertex, and check for the minimum weighted outgoing edge

Here, it's A→D

(for undirected graph, er assumed edges from A to be outgoing)

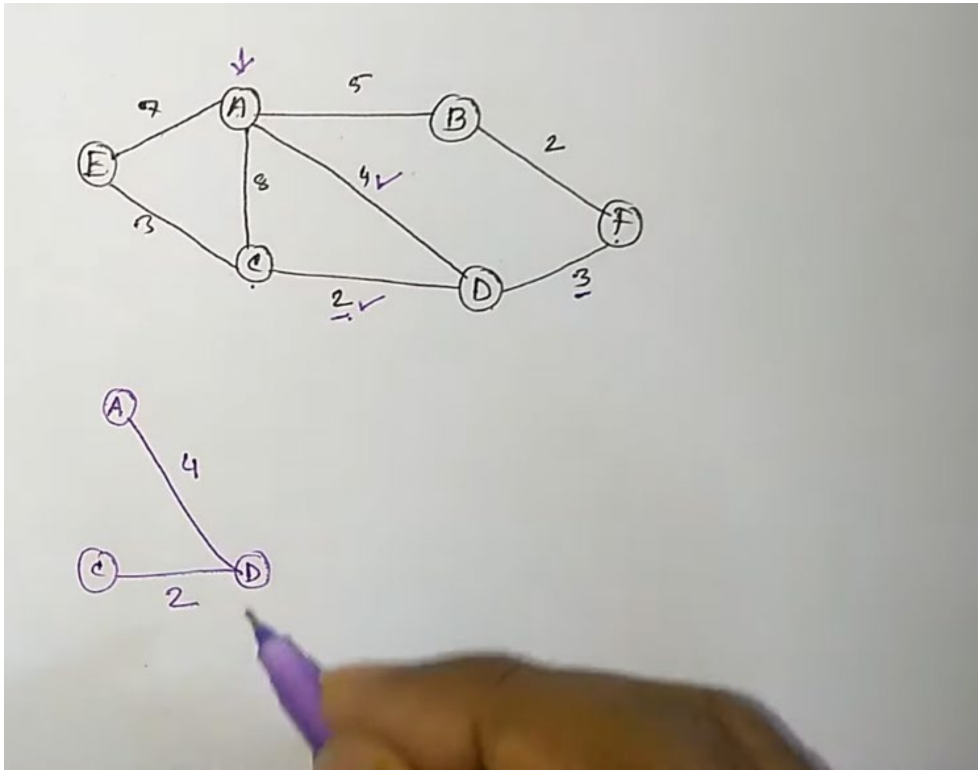


Step 2

For each step, we get a new vertex from the algorithm. But unlike Kruskal’s Algorithm, we now need to check the outgoing edges from both A and D.

Meaning we can continue adding edges from either A or D.

Here, the minimum edge around A and D is C, therefore:



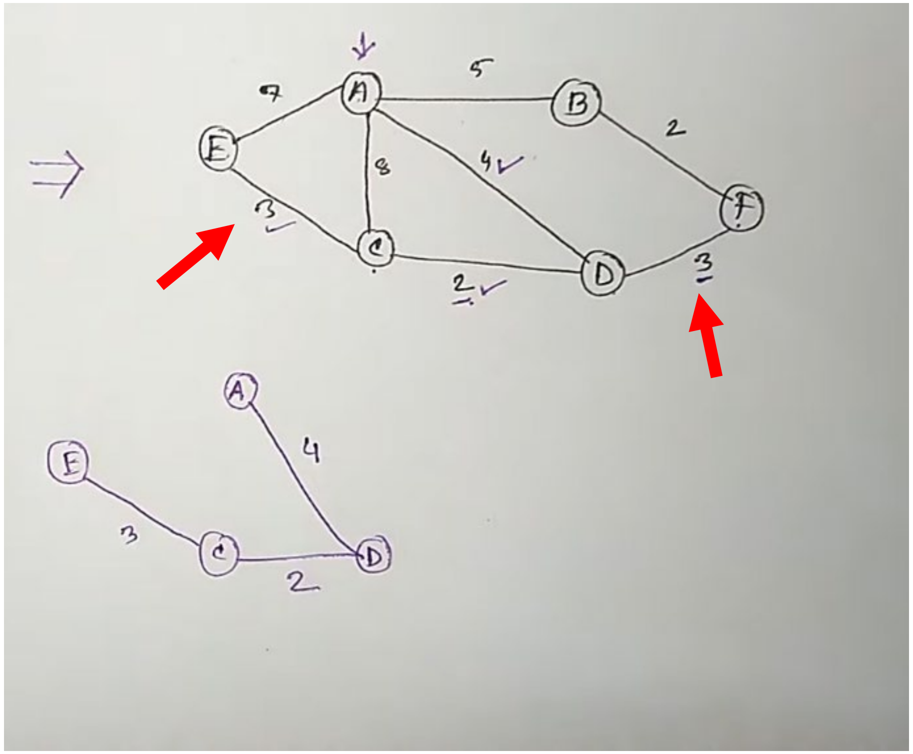
Step 3

When there are equal minimum edges, like in this case,

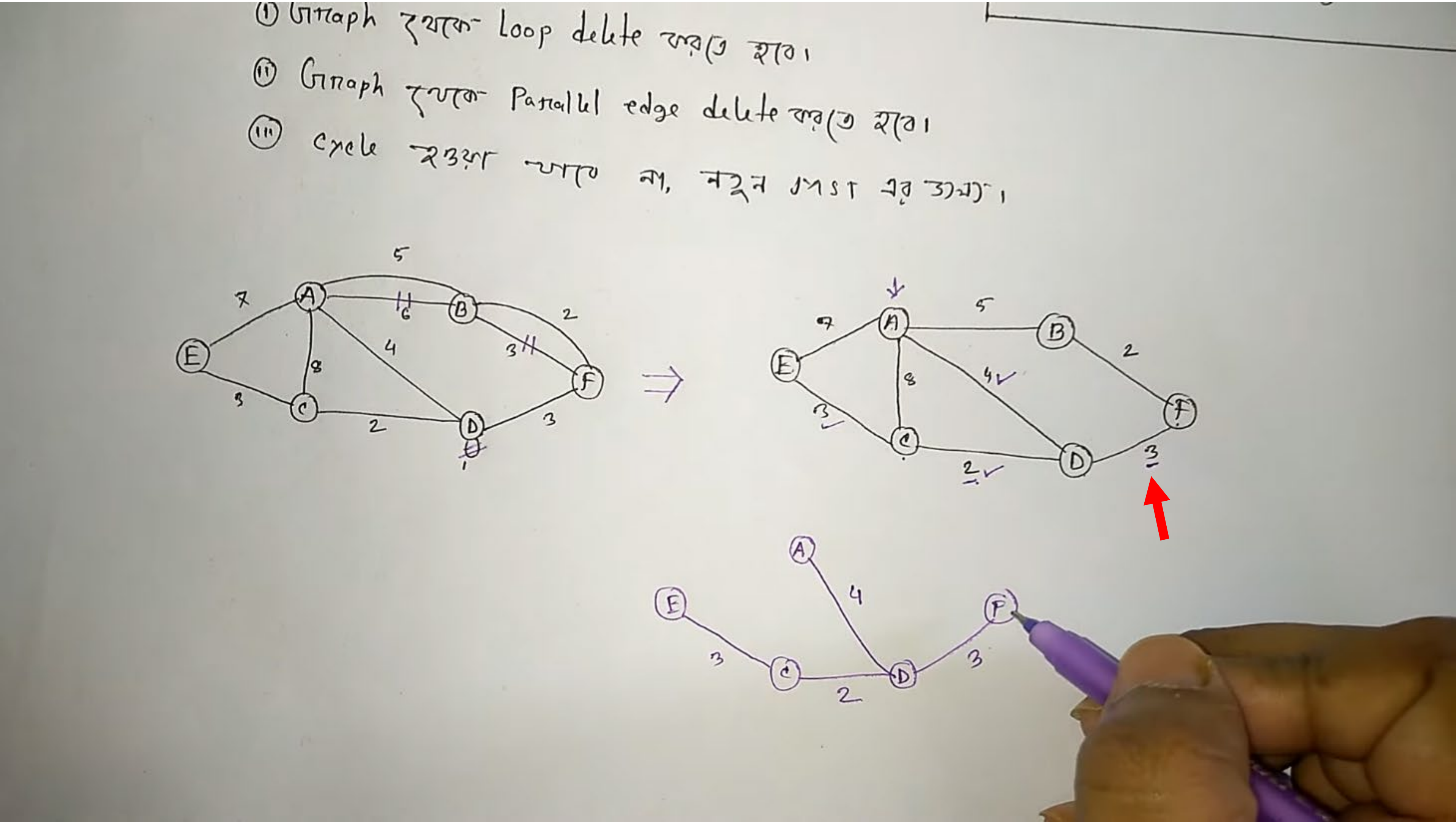
C → E

D → F

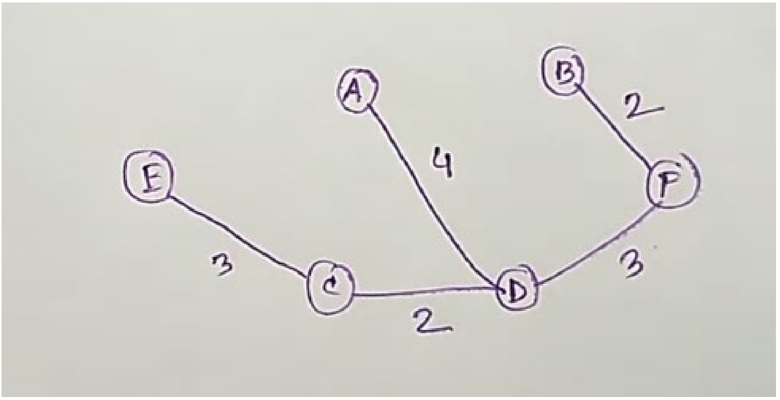
We can take any one of them, it does not matter.



Step 4

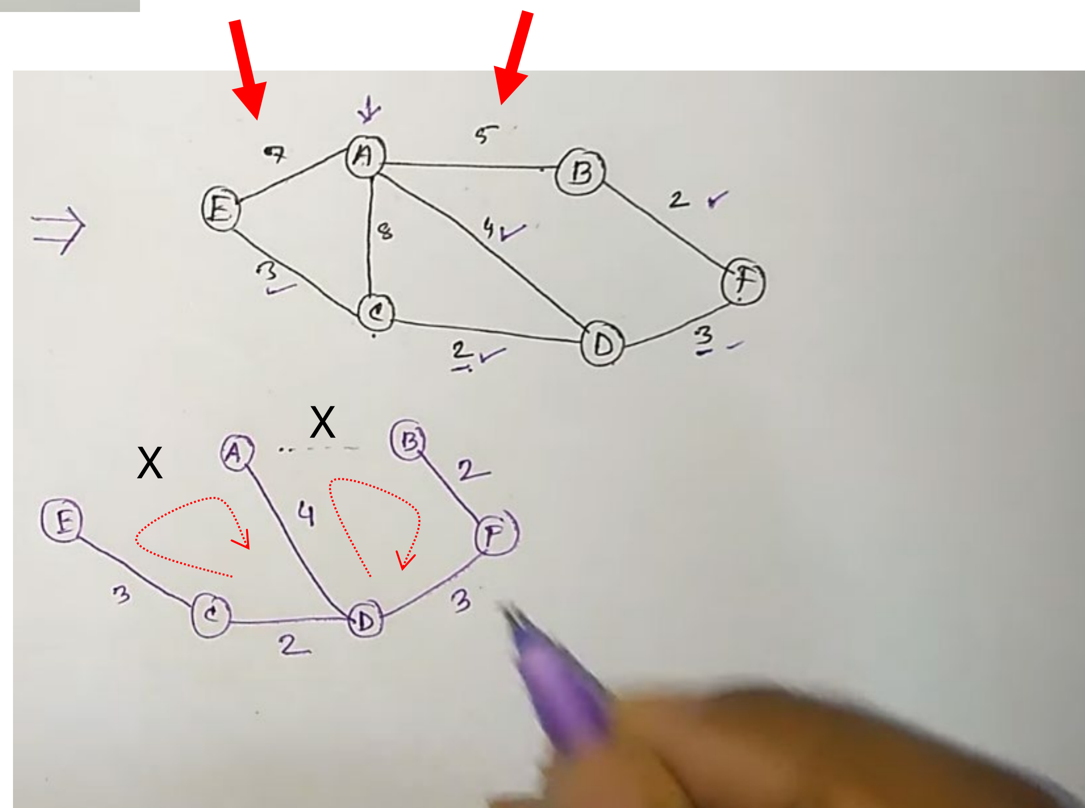

(i) Graph হতে Loop delete করতে হবে।
(ii) Graph হতে Parallel edge delete করতে হবে।
(iii) Cycle হওয়া যাবে না, নতুন MST এর জন্য।

Step 5



Step 6

Taking any more edges will create a cycle, therefore, this is the MST using Prim’s Algorithm



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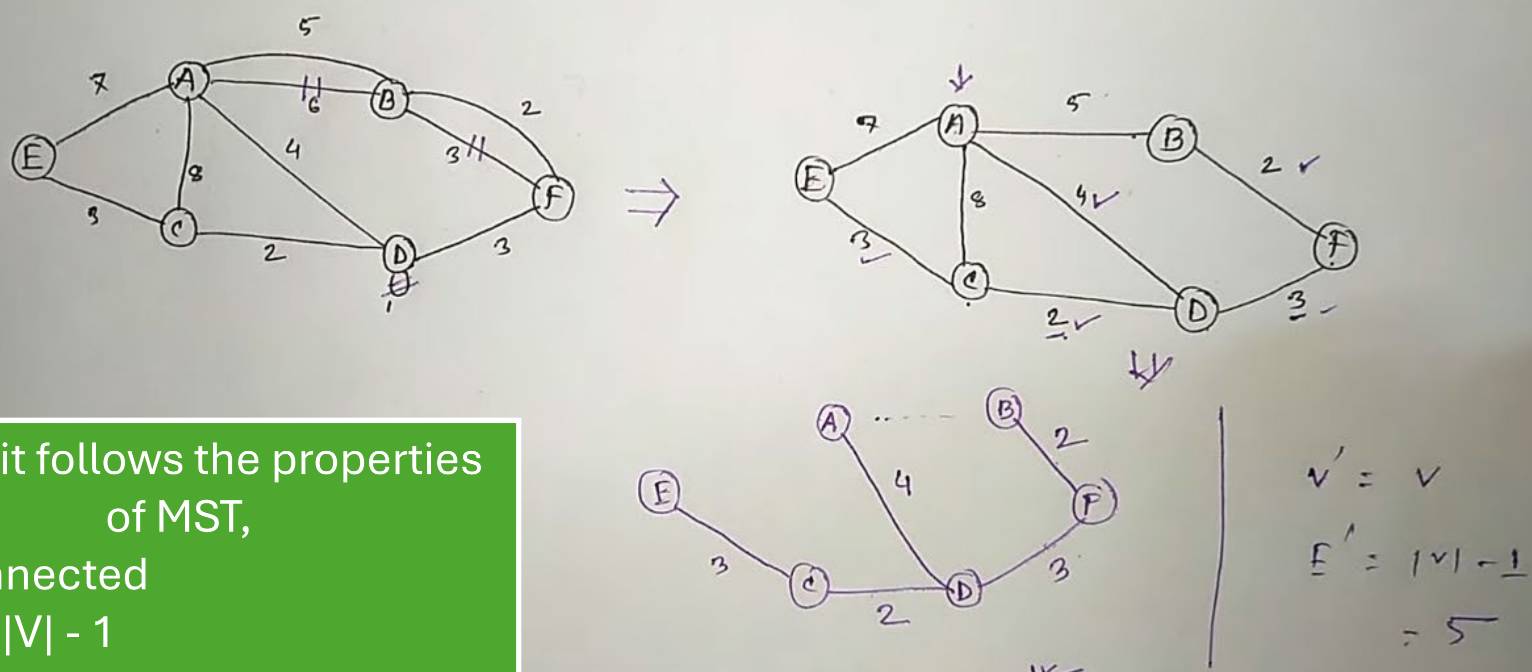

Since it follows the properties of MST,

- Connected
- $E' = |V| - 1$
- $V = V'$

This is a valid MST